

VARUN SHARMA

AI Engineer | Technical Lead | Agentic AI, LLM Platforms & Cloud Architecture
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PROFESSIONAL SUMMARY

AI engineer and technical lead who designs and ships production LLM systems for enterprise clients. Own architecture decisions, code review standards and production release sign-off across concurrent client deliveries. Work spans the full lifecycle: agentic AI and multi-agent systems (LangGraph, Model Context Protocol), retrieval-augmented generation, AutoML for tabular workloads, multi-tenant FastAPI backends, and the MLOps, cloud and AI observability layers that keep LLM features reliable in production.

CORE COMPETENCIES

Generative AI & Agents: Agentic AI, AI agents, multi-agent systems, LangGraph, Model Context Protocol (MCP), RAG (hybrid and semantic retrieval), agent memory, prompt engineering, LLM evaluation, AI observability

Machine Learning & MLOps: AutoML, forecasting, classification, regression, clustering, anomaly detection, feature engineering, SHAP explainability, model deployment, training pipelines, approval-gated model promotion

Platform & Cloud Architecture: Multi-tenant SaaS, distributed systems, event-driven design, RBAC/ABAC, API design, caching strategy, Kubernetes, Terraform, CI/CD, cloud cost and performance tuning

Technical Leadership: Team leadership, architecture ownership, code review standards, sprint planning, release sign-off, client-facing technical consulting, mentoring

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, Bash

AI / LLM: LangChain, LangGraph, DSPy, LiteLLM, Model Context Protocol (MCP), multi-agent orchestration, RAG, LLM evaluation, Langfuse

Models: OpenAI GPT, Anthropic Claude, Google Gemini, Llama, Qwen, Mistral (open-weight serving via vLLM and Ollama)

Machine Learning: scikit-learn, XGBoost, SHAP, AutoML, feature engineering, forecasting, anomaly detection

Backend: FastAPI, Node.js, Express, Django, REST, GraphQL, WebSockets, microservices, Celery, RabbitMQ

Data Stores: PostgreSQL, MongoDB, Redis, Firestore; Weaviate, Qdrant, ChromaDB, FAISS (vector databases)

Cloud & DevOps: GCP (Cloud Run, Cloud Functions), AWS (EC2, S3, Lambda, ECS), Azure, DigitalOcean, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Nginx, Linux

Observability: Prometheus, Grafana, OpenTelemetry, Langfuse, ELK

Frontend: React, Next.js, SvelteKit, Tailwind CSS, Puppeteer

PROFESSIONAL EXPERIENCE

NerveSparks Pvt. Ltd. - New Delhi, India

Nov 2024 - Present

Technical Lead - AI Engineering

Leadership & Delivery Ownership

- Lead a cross-functional engineering team delivering enterprise AI products to multiple concurrent clients; own architecture decisions, code review standards, sprint planning and production release sign-off.
- Act as technical point of contact in client discussions, translating business requirements into system designs, scope and delivery plans.
- Mentor junior engineers through architecture reviews, pair programming and technical planning.

Generative AI & Agentic Systems

- Designed agentic AI workflows using LangGraph and custom MCP servers, enabling tool-calling agents to orchestrate retrieval, external APIs and multi-step reasoning behind a single service interface.
- Built a centralised RAG service on FastAPI, LangChain and Weaviate with a provider-agnostic LLM abstraction (OpenAI, Anthropic Claude, Google Gemini, open-weight models), reused across several client applications.
- Implemented long-term agent memory on vector retrieval, plus evaluation pipelines for retrieval quality and hallucination detection.
- Engineered a meeting-agent service using Puppeteer to join Google Meet, Teams and Zoom, capture audio in real time and produce searchable LLM summaries stored as semantic memory.

- Built a real-time conversational AI avatar combining speech recognition, LLM reasoning and speech synthesis for customer-facing interaction.

Machine Learning & MLOps

- Led an AutoML platform supporting forecasting, classification, regression, clustering and anomaly detection, with automated feature engineering, model selection and SHAP-based explainability.
- Built configurable ML data pipelines covering profiling, preprocessing and training runs, with approval workflows gating model promotion to production.

Platform Engineering, Cloud & Reliability

- Architected a multi-tenant enterprise AI platform with hierarchical RBAC (Platform, Organisation, Department, User) on Google Identity Platform, including tenant-level data isolation and audit trails.
- Shipped 15+ FastAPI microservices to Cloud Run and Kubernetes with autoscaling, cold-start tuning and zero-downtime deploys.
- Standardised infrastructure with Terraform across AWS, GCP, Azure and DigitalOcean, with Nginx load balancing fronting 10+ services.
- Cut deployment time by roughly 60% by rebuilding CI/CD on GitHub Actions and Jenkins with automated test, build and rollback stages.
- Reduced P95 API latency by roughly 40% through query profiling, indexing and a three-tier cache (in-memory, Redis, CDN) sustaining an ~85% hit ratio.
- Established the observability stack (Prometheus, Grafana, OpenTelemetry, Langfuse) covering infrastructure metrics alongside LLM traces and cost, cutting MTTR by roughly 50%.

Manav Rachna University - Faridabad, India

Aug 2024 - Oct 2024

Web Developer Intern

- Built the Student Welfare Department portal, centralising information access for 5,000+ students and faculty.
- Developed real-time feedback modules for student-faculty communication.
- Supported data analysis and visualisation for an HRV biomedical research project.

FEATURED PROJECTS

Enterprise AI Platform

FastAPI, LangGraph, MCP, Weaviate, PostgreSQL, Redis, Docker, Cloud Run

- Unified abstraction layer over multiple LLM providers with per-tenant API key management, routing and cost tracking.
- Long-term agent memory built on vector retrieval; evaluation pipelines for retrieval quality and hallucination detection reached 92% accuracy on internal test sets.

AutoML & ML Pipeline Service

Python, FastAPI, scikit-learn, XGBoost, SHAP, PostgreSQL

- Config-driven pipeline handling profiling, feature engineering, model selection and deployment for tabular workloads.
- SHAP explainability surfaced per-prediction drivers so non-technical stakeholders could sign off on model behaviour.

AI Document Processing System

Python, LangChain, DSPy, ChromaDB, FastAPI

- Modular ingestion for PDF, DOCX and image sources with semantic chunking and metadata-aware retrieval.
- DSPy-based automatic prompt optimisation improved answer relevance by roughly 35% over hand-written prompts.

Imaginify - AI Image Generation Platform

React, Node.js, MongoDB, Redis, Hugging Face, Stripe

- Text-to-image product on Stable Diffusion with credit-based Stripe billing, Redis caching and a job queue for concurrent generation.

EDUCATION

Master of Computer Applications - Manav Rachna International University

Jul 2023 - Jun 2025

Faridabad, India | GPA 8.1 / 10

Bachelor of Computer Applications - Maharshi Dayanand University

Jul 2020 - May 2023

Rohtak, India | GPA 7.2 / 10